

Instructions & how to succeed in the Digital Signal Processing course

Andreas Austeng^a and Sven Peter Näsholm^b

*Department of Informatics,
University of Oslo*

(Dated: 18 August 2025)

Welcome to the wonderful world of Digital Signal Processing! We provide a broad set of sessions that will help you develop your digital signal processing skills.

We highly recommend you to make use of these offerings – experience from previous years tells that those who use these services have greater success.

Do not hesitate to ask for help if you get stuck!

CONTENTS

I. How to qualify for the exam	1
II. Weekly schedule	2
III. Electronic forum	3
IV. Code examples and interactive resources / tutorials	3
V. Exam	4
VI. Some pieces of advice	4
VII. An international and national context	4
A. Resources and video lecture links	4
B. Digital signal processing course contents with other universities	4
A. Weekly time spent – approximate add-up	6

I. HOW TO QUALIFY FOR THE EXAM

Both requirements 1) and 2) below must be fulfilled in order for you to be allowed to take the final exam:

1) Project

- You will be provided with the project task around halfway into the course and it will take you around one week to finish it.
- Your report and code from the Project must be approved for you to be eligible to take the exam.

2) Weekly delivery tasks

- We provide 12 “weekly delivery tasks”. Each set contains tasks with points summing up to more than 10 points.
- You can gain 10 points each week, which means that you do not have to succeed (or submit) on all of the tasks in order to gain 100% of the score.
- Requirements to be eligible to take the final course exam:
 - Bachelor students: you must collect 48 out of the total 120 points in order to be eligible to take the exam.

^a andrea@ifi.uio.no

^b svenpn@ifi.uio.no

- Master students: you must collect 72 out of the total 120 points in order to be eligible to take the exam.
- Procedure during the Friday submission and correction sessions:
 - *Your presence is mandatory on one of the the Friday submission and correction sessions* (for all weeks you want to collect points)
 - If you're sick or have special constraints for example due to a part-time job: send an application – including motivation – to the course responsables and you might be allowed to submit the exercises electronically before the first Friday session starts in that week.
 - The submissions (which must be on paper) are collected in the beginning of the session. These are then randomly distributed among the students for peer-correction.
 - The group teacher will highlight important aspects of the solutions of this week's task.
 - For each task: write the score given. The group teachers will tell what is an adequate quantization of points (typically the quantization step is 0.5 points)
 - Also write on the submission: the name of the peer that has corrected it.
 - The corrected submission is delivered to the group teacher who will check the corrections, or samples of them, and who has the final say about the scoring.
 - Scores are registered in the *Devilry* system: <https://devilry.ifi.uio.no/>

II. WEEKLY SCHEDULE

	Monday	Tuesday	Wednesday	Wednesday	Friday (2 sessions)
Week i	Video intro $i + 1$ Elaboration lecture i	Group lecture i	Discussion lecture i	Group lecture i	Exercise delivery i
Week $i + 1$	Video intro $i + 2$ Elaboration lecture $i + 1$	Group lecture $i + 1$	Discussion lecture $i + 1$	Group lecture $i + 1$	Exercise delivery $i + 1$

We offer a video intro plus a total of 8h live teaching and support per week, in addition to the weekly mandatory task delivery sessions on Friday.

1) Intro video

Published: The week before

Contents: Video with draft introduction to the topics of the week.

NOTE:

- The video is an *introduction* and does not provide the full-fledged teaching of the topics of the week.
- The live sessions on-site *expect you to have seen the introduction video* before you come.

2) Discussion / elaboration / questions-and-answers session 1

Time: Mondays 12h15–14h

Contents: Highlights and explanations of important aspects of the week's topics. Discussions and explanations based on students' recommendation

Expectation: We assume that you have watched the video before you come to this session.

Highlight: Often also providing hints, or even full solutions, to parts of the weekly exercises!

Recommendation: Be active in order to stimulate dialogue and discussions for your own and your peers' development!

In weeks with guests: Guests displaying their use of digital signal processing in their jobs.

3) Discussion / elaboration / questions-and-answers session 2

Time: Wednesdays 8h15–10h

Contents: In weeks without guests: Continuation of Discussion & elaboration, sometimes complemented with Q&A session.

In weeks with guests: Guests displaying their use of digital signal processing in their jobs.

4) Group A assistance session Plenary exercises / Regneøvelser

Teacher: Krithika or Brage

Time: Tuesdays 08h15–10h

Contents: The group teacher helps you to get started solving the weekly deliveries.

Recommendation: Be sure to ask for help if you get stuck

Remark: You're welcome to join both group assistance sessions

5) Group B assistance session Plenary exercises / Regneøvelser

Teacher: Krithika or Brage

Time: Wednesdays 10h15–12h

Recommendation: Be sure to ask for help if you get stuck

Remark: You're welcome to join both group assistance sessions

6) Weekly mandatory delivery session, group A

Teacher: Krithika or Brage

Time: Fridays 08h15–12h

Content: Peer-correction and submission of weekly mandatory deliveries.

Presence: *Mandatory*

Process: You will be tasked to correct one of your peers' deliveries, assisted by the solution suggestions provided by the group teacher. Write a remark if you're unsure about the correction. The group teachers will collect and will check the scoring of a subset of the deliveries.

Score follow-up: The *Devilry* system will be used for the registration of scoring of each student.

7) Weekly mandatory delivery session, group B

Teacher: Krithika or Brage

Time: Fridays 10h15–12h (same conditions as for group A)

III. ELECTRONIC FORUM

We will use the *Discourse* for dialogue between students and with lecturers / group teachers:

- URL: <https://astro-discourse.uio.no/c/in3190-25h>

This will serve as a questions-and-answers database for the benefit of all students.

We expect everybody to maintain a professional and polite tone in the forum.

Please login using the link above and your UiO credentials.

IV. CODE EXAMPLES AND INTERACTIVE RESOURCES / TUTORIALS

There is a course GitHub repository available where you find several tutorials, examples, and coding snippets that will be useful. There is material both in Matlab and Python:

- <https://github.uio.no/ifidsb/IN3190>

V. EXAM

- Oral exam in the end of the semester.
- No books or notes allowed during the exam.
- No extra exam will be offered to students who have withdrawn from the exam, or who did not pass.

VI. SOME PIECES OF ADVICE

- Work together and learn from each other.
- Be active in lectures and group sessions.
- Don't hesitate to seek help when you're about to get stuck.
- Do you find a particular topic difficult? Ask the lecturer for an additional session to explain.
- The needs can vary between students, in particular because the background vary greatly. Some might initially struggle a bit with the mathematics, while others have less experience in coding or in working with data recordings. It is challenging to satisfy the full multi-background audience all the time. But we make honest attempts!
- Please see Appendix A for a suggested flowchart and an illustration of how the weekly time spent can add up.

VII. AN INTERNATIONAL AND NATIONAL CONTEXT

A. Resources and video lecture links

Digital Signal Processing is an integrated component in engineering and science university educations worldwide. These fundamental theories and methods have remained relevant for several decades now and have become even more appropriate in our modern era of data centered industries, machine learning, and computationally intensive research and innovation.

There are several electronic teaching resources available via the world wide web, for example:

- Very nice and thorough video lectures by John Buck, University of Massachusetts [his homepage is <https://www.umassd.edu/directory/jbuck/>]. Videos: <https://www.youtube.com/user/ProfJohnBuck/videos> – see the *Discrete Time Linear Systems* playlist
- MIT material by the legendary Alan V. Oppenheim: <https://ocw.mit.edu/courses/res-6-008-digital-signal-processing-spring-2011/>
- Even more Oppenheim lecturing video material <https://www.youtube.com/playlist?list=PL8157CA8884571BA2>
- ... as well as several other online outlets

B. Digital signal processing course contents with other universities

The contents of digital signal processing courses are quite standardized. Examples:

University of Tromsø: Discrete-time signal and systems, z-transform, discrete-Fourier transform, fast-Fourier transform, sampling, discretization, frequency response, impulse response, filter design techniques, correlation, convolution, interpolation, decimation.

NTNU: Time, frequency, and z-domain description of discrete signals and linear time-invariant systems; analysis and design of filters; multirate systems, correlation, and energy spectrum; statistical properties, modeling and estimation of discrete stochastic processes.

DTU Copenhagen (5p course): Time-domain analysis and difference equations / Sampling theorem / Discrete-time Fourier transform (DTFT) / Fast Fourier transforms (FFT) / z-transform / IIR and FIR digital filters / The short-time Fourier transform / Spectral density of a stochastic signal / Noise reduction

KTH Stockholm: The course address classical results and methods in digital signal processing, both in terms of fundamental principles and mathematical theories. In particular: Implementation of digital filters and approximations of target impulse responses and transfer functions using FIR filters. / Spectral estimation methods including the periodogram, the modified periodogram, and auto regressive (AR) model based spectral estimation methods and their connection to prediction. / The FFT algorithm and its use in spectral estimation and filtering. / Upsampling and downsampling of time-discrete signals. / Filterbanks for splitting a signal into sub-bands. / Effects of quantization and fixed point implementations of filters and systems.

ETH Zürich (excluding the statistical signal processing part of their extended course): Discrete-time linear systems and filters: state-space realizations, z-transform and spectrum, decimation and interpolation, digital filter design, stable realizations and robust inversion. The discrete Fourier transform and its use for digital filtering.

Caltech: Introduction to continuous and discrete time signals and systems with emphasis on digital signal processing systems. Study of the Fourier transform, Fourier series, z-transforms, and the fast Fourier transform as applied in electrical engineering. Sampling theorems for continuous to discrete-time conversion. Difference equations for digital signal processing systems, digital system realizations with block diagrams, analysis of transient and steady state responses, and connections to other areas in science and engineering.

DIGITAL SIGNAL PROCESSING "COURSE WEEKLY FLOWCHART"

